Supplementary Information

Document caption: Detailed methods, supplementary analyses, and results including p-values.

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1 Current climate conditions

Data is available 2007 – 2022. 2008 is the first whole year of temperature data. Data available from https://data.g-e-m.dk (GEM 2026f, 2026d).

1-1 Tempterature

Monthly mean temeprature

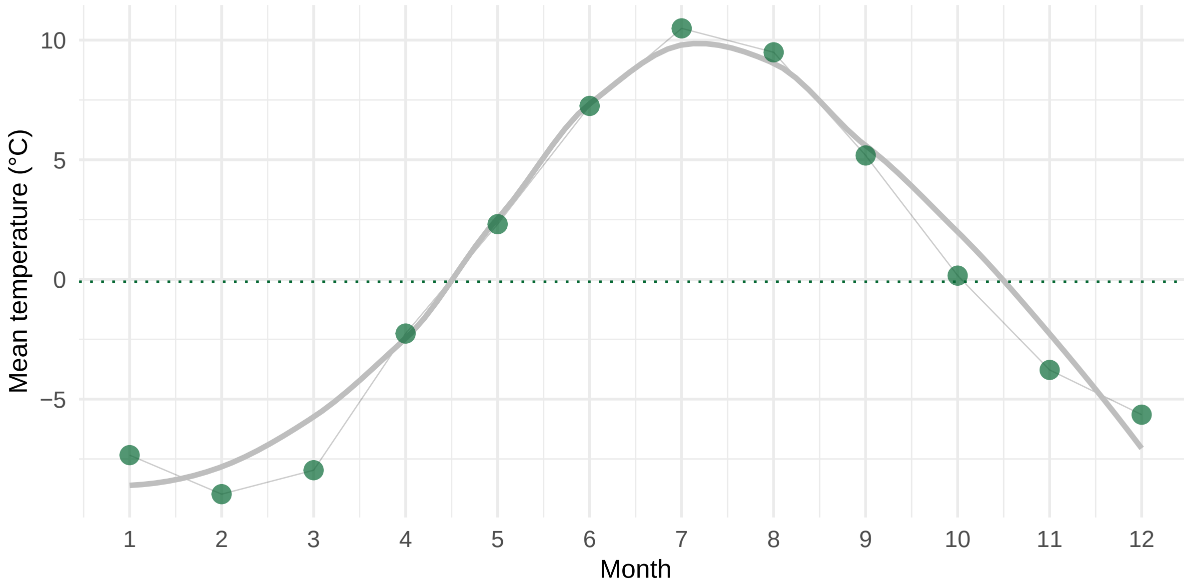

Figure 1: Mean temperature pr month. Mean air temperature (°C) pr month, based on data from 2007 to 2022. The dashed line represent yearly mean of -0.1°C.

Yearly mean temperature

```r
model_temp_annual_mean <- lm(mean ~ year, data = data_temp_annual_mean)
summary(model_temp_annual_mean)
```

```
Call:
lm(formula = mean ~ year, data = data_temp_annual_mean)

Residuals:
    Min      1Q  Median      3Q     Max 
-2.4656 -0.6153 -0.1082  0.6452  3.3959 

Coefficients:
              Estimate Std. Error t value Pr(>|t|)
(Intercept)   7.679175 170.533389   0.045    0.965
year         -0.003821   0.084632  -0.045    0.965

Residual standard error: 1.416 on 13 degrees of freedom
Multiple R-squared:  0.0001568, Adjusted R-squared:  -0.07675 
F-statistic: 0.002038 on 1 and 13 DF,  p-value: 0.9647
```

```
data_temp_annual_mean$pred <- predict(model_temp_annual_mean)
```

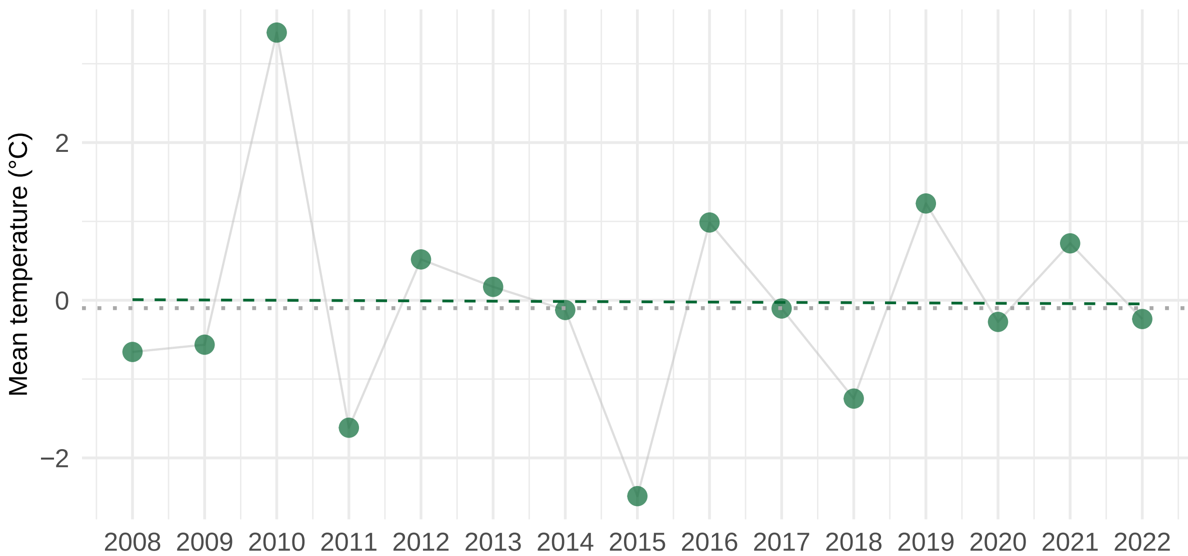

Figure 2: Mean annual air temperature pr year (°C) from 2008 to 2022 in Kangerluassunguaq. Data from 2007 is excluded because data is only from October, November and December. Gray dotted line indicate over all mean of -0.1003°C. Dashed line is model prediction.

Five year intervals of temperature

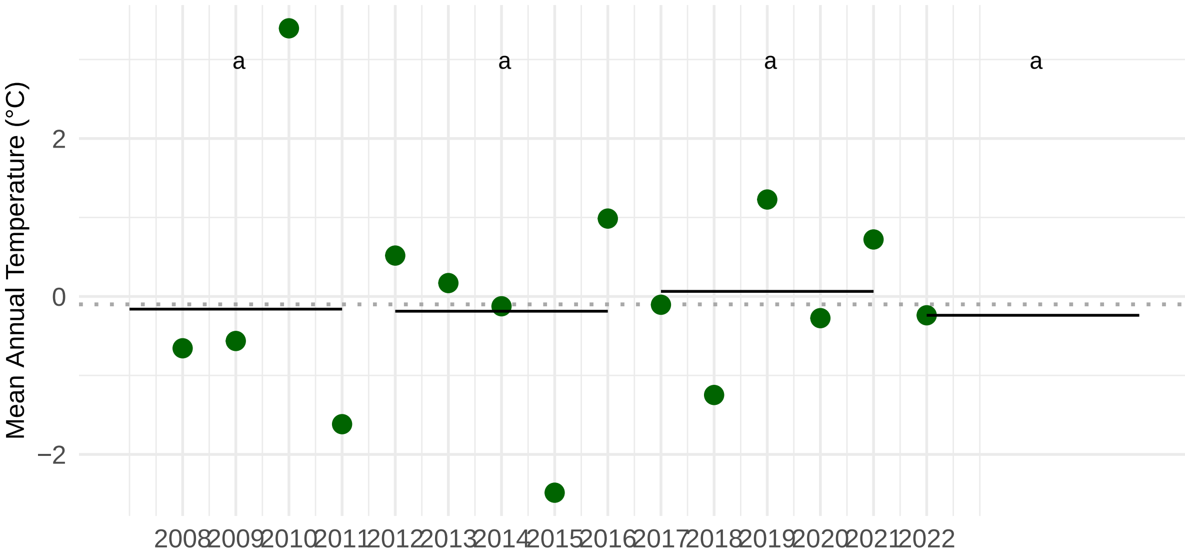

Figure 3: Five year intervals in temperature means. Shared letters indicate non-significant differences among estimated means according to compact letter display.

1-2 Precipitation

Data is available from GEM. (2020). *ClimateBasis Nuuk—Precipitation—Precipitation accumulated (mm)* (Version 1.0) [Dataset]. Greenland Ecosystem Monitoring. https://doi.org/10.17897/SXJ8-WA79. Downloaded on XXXX.

The mean annual precipitation is 886.15 mm.

The months with the most precipitation are September (138.64 mm), October (99.38 mm), August (98.28 mm), which account for 336.3 mm or 38 % of the yearly precipitation.

The months with the least precipitation are May (51.51 mm), February (37.62 mm), March (36.44 mm).

Monthly mean precipitation

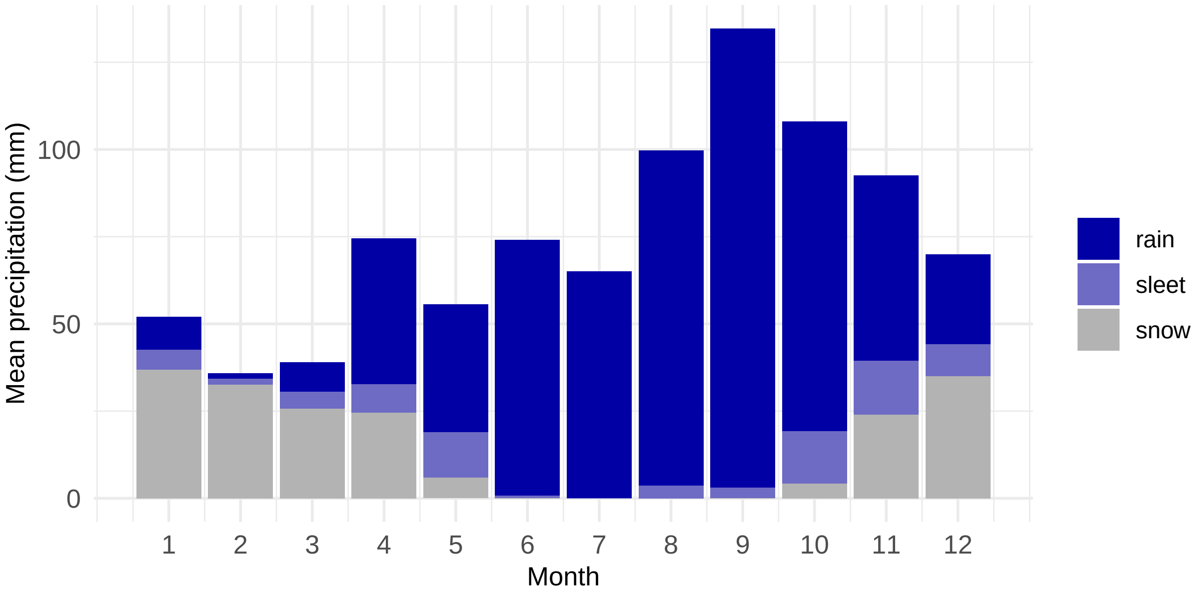

Figure 4: Mean precipitation type pr month. Mean monthly distribution of types of precipitation (mm). Based on data from 2008 to 2022 (requires join with temperature data which it only available till 2022). Sleet is defined as precipitation that fell when temperatures were between -1°C and 1°C.

Yearly mean precipitation

```
Call:
lm(formula = yearly_precip ~ year, data = data_precip_mean_annual)

Residuals:
    Min      1Q  Median      3Q     Max
-405.29 -221.91   54.49  205.78  307.56

Coefficients:
             Estimate Std. Error t value Pr(>|t|)
(Intercept) -9763.076  24027.287  -0.406    0.690
year            5.282     11.918   0.443    0.664

Residual standard error: 240.7 on 15 degrees of freedom
Multiple R-squared:  0.01293,   Adjusted R-squared:  -0.05288
F-statistic: 0.1964 on 1 and 15 DF,  p-value: 0.6639
```

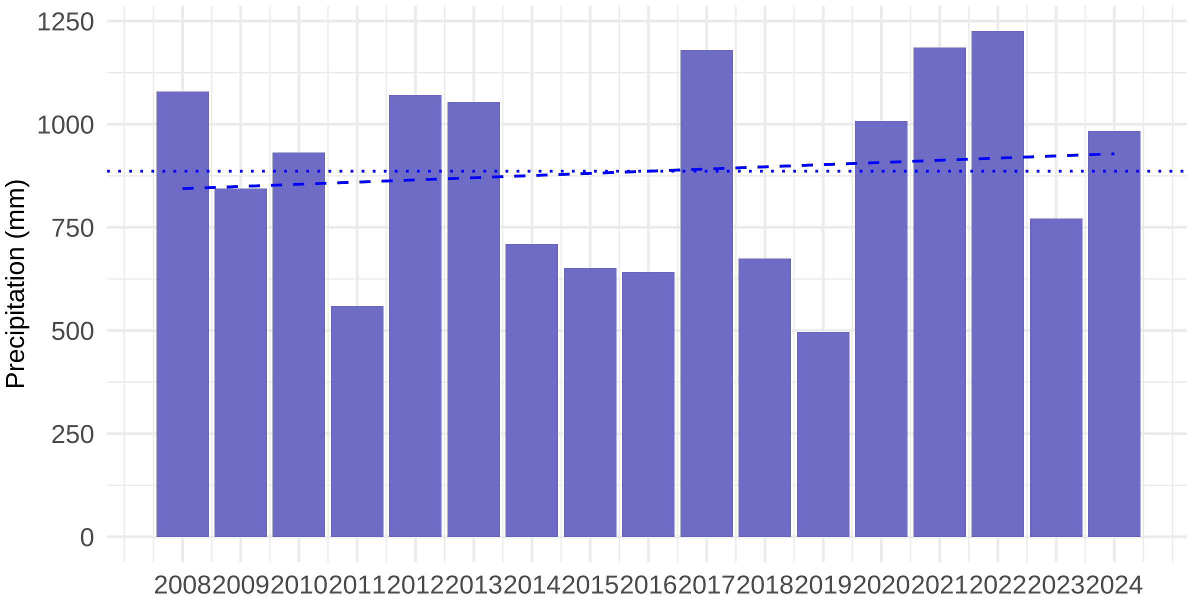

Figure 5: Mean yearly precipitation (mm) from 2008 to 2024. Data from 2007 is excluded because it was only from May - December. Blue dashed line indicate over all mean of 886.15 mm. Dashed line represent model prediction.

Five year interval precipitation

```
      date                   time                       pre_mm
 Min.   :2008-01-01   Min.   :00:00:00.000000   Min.   : 0.0000
 1st Qu.:2012-07-23   1st Qu.:06:00:00.000000   1st Qu.: 0.0000
 Median :2016-09-30   Median :12:00:00.000000   Median : 0.0000
 Mean   :2016-08-30   Mean   :11:30:21.122933   Mean   : 0.1058
 3rd Qu.:2020-12-09   3rd Qu.:18:00:00.000000   3rd Qu.: 0.0000
 Max.   :2024-12-31   Max.   :23:00:00.000000   Max.   :19.0000
 quality_flag             year          month             doy
 Length:142395      Min.   :2008   Min.   : 1.000   Min.   :  1.0
 Class :character   1st Qu.:2012   1st Qu.: 4.000   1st Qu.: 99.0
 Mode  :character   Median :2016   Median : 7.000   Median :187.0
                    Mean   :2016   Mean   : 6.625   Mean   :186.3
                    3rd Qu.:2020   3rd Qu.:10.000   3rd Qu.:276.0
                    Max.   :2024   Max.   :12.000   Max.   :366.0
 month_string           anchor       seg_start       seg_end
 Length:142395      Min.   :2007   Min.   :2007   Min.   :2011
 Class :character   1st Qu.:2007   1st Qu.:2012   1st Qu.:2016
 Mode  :character   Median :2007   Median :2012   Median :2016
```

```
                    Mean   :2007   Mean   :2014   Mean   :2018
                    3rd Qu.:2007   3rd Qu.:2017   3rd Qu.:2021
                    Max.   :2007   Max.   :2022   Max.   :2026
   segment          seg_mean_precip
 Length:142395      Min.   :2980
 Class :character   1st Qu.:3413
 Mode  :character   Median :4128
                    Mean   :3877
                    3rd Qu.:4544
                    Max.   :4544
```

1-3 Radiation

Data available from https://data.g-e-m.dk (GEM 2026e)

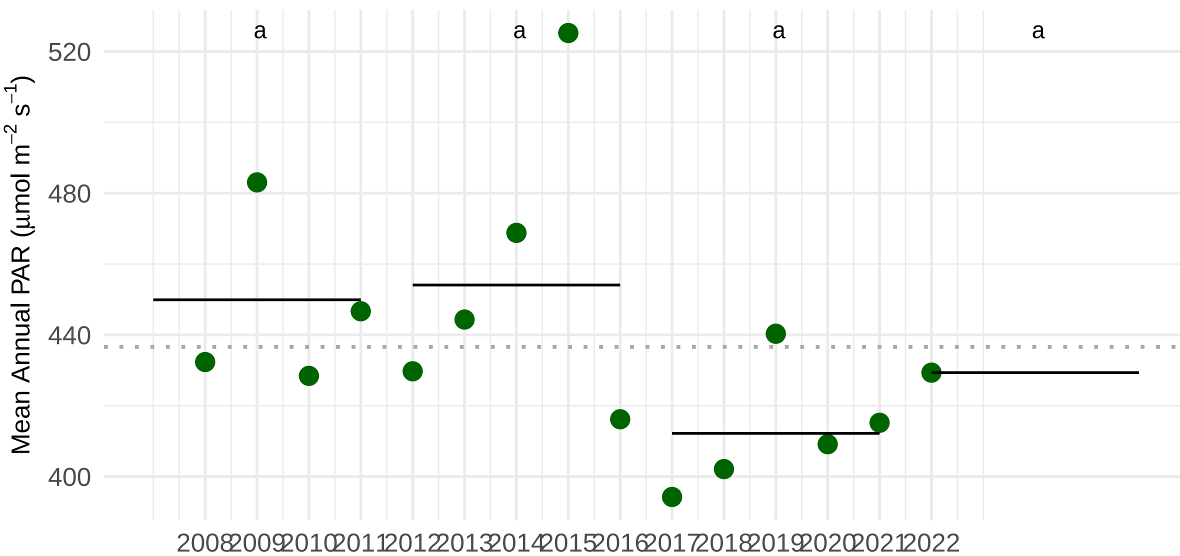

Figure 6: Five year intervals in PAR means. Shared letters indicate non-significant differences among estimated means according to compact letter display.

2 Historic climate data

2-1 Temperature

Data available from Cappelen (2021).

Station 4250 is Nuuk, and element 101 is precipitation.

```r
model_annual_temp_historic <- lm(annual ~ year, data = nuuk_21_04_dmi_temp)
summary(model_annual_temp_historic)
```

```
Call:
lm(formula = annual ~ year, data = nuuk_21_04_dmi_temp)

Residuals:
    Min      1Q  Median      3Q     Max 
-3.3697 -0.6837  0.0834  0.7403  3.3848 

Coefficients:
              Estimate Std. Error t value Pr(>|t|)    
(Intercept) -19.764947   2.851721  -6.931 8.06e-11 ***
year          0.009443   0.001478   6.389 1.51e-09 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 1.163 on 172 degrees of freedom
Multiple R-squared:  0.1918,	Adjusted R-squared:  0.1871 
F-statistic: 40.81 on 1 and 172 DF,  p-value: 1.512e-09
```

```r
nuuk_21_04_dmi_temp$pred <- predict(model_annual_temp_historic)

preds <- predict(model_annual_temp_historic, se.fit = TRUE)

# 2. Create a new dataframe for the ribbon
# We calculate the Upper and Lower bounds (Fit ± 1.96 * SE for 95% CI)
nuuk_21_04_dmi_temp$fit <- preds$fit
nuuk_21_04_dmi_temp$se <- preds$se.fit
nuuk_21_04_dmi_temp$lower <- preds$fit - 1.96 * preds$se.fit
nuuk_21_04_dmi_temp$upper <- preds$fit + 1.96 * preds$se.fit
```

```r
ggplot(data = nuuk_21_04_dmi_temp, aes(x = year, y = annual)) +
    geom_line(
    aes(y = pred),
    color = "#076834",
    linetype = "solid",
  ) +
    geom_ribbon(aes(ymin = lower, ymax = upper),
              fill = "#076834",
```

```
             alpha = 0.2) +
  geom_point(size = 3, alpha = 0.7, color = "#076834") +
  #geom_smooth(method = "lm")+
  scale_x_continuous(
    breaks = seq(min(gr_monthly_all_1784_2020$year),
                 max(2025), by = 15))+
  labs(x = "", y = "Temperature (°C)") +
  theme_minimal()+
    theme(  axis.title.x = element_text(size = 10),
  axis.title.y = element_text(size = 10),
  axis.text.x  = element_text(size = 10),
  axis.text.y  = element_text(size = 10))
```

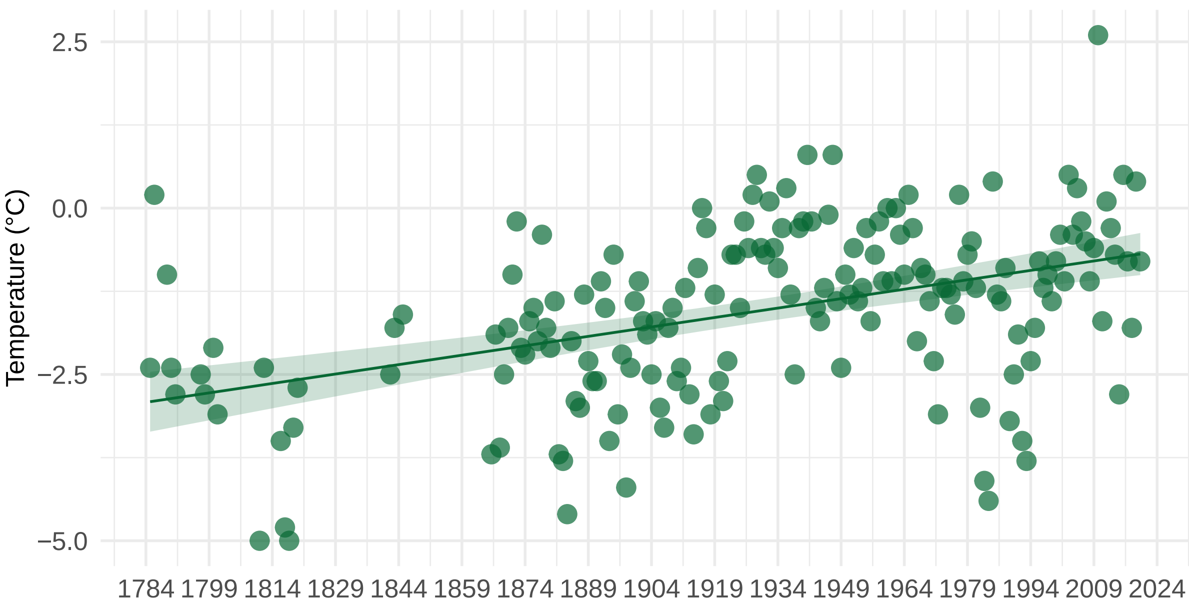

Figure 7: Historic yearly mean air temperature (°C) in Nuuk, from 1785 to 2020. Yearly mean of -1.6°C. Yearly mean ranges from -5°C to 2.6°C. Modelled mean temperature has increase from -2.92°C in 1784 to -0.65°C in 2024 (increase of 2.27°C or 0.095°C decade [1].)

3 Statistical analysis

3-1 Diversity

Data available from https://data.g-e-m.dk (GEM 2026a).

Importing data and calculating indexes

```
#reading the raw datafile
data_raw <- readRDS("~/Library/CloudStorage/OneDrive-Aarhusuniversitet/MappingPlants/01 Vege

data_filtered <- data_raw |>
  group_by(year, subsection) |>
  mutate(no_plots = n_distinct(plot_id)) |>
  ungroup() |>
  filter(taxon_code != "rock", no_plots > 1, veg_type != "saltmarsh") |>
  dplyr::select(year,vt_section,subsection, plot_id, no_plots, veg_type, taxon_code, presenc

data_pivotwide <- data_filtered |>
  tidyr::pivot_wider(
    id_cols = c(year, vt_section, subsection, plot_id, no_plots, veg_type),
    names_from = taxon_code,
    values_from = presence,
    values_fill = 0,
    values_fn = max
  )

data_pivotlong <- data_pivotwide |>
  pivot_longer(cols = carbig:viospe, names_to = "taxon_code", values_to = "presence")

#calculation count and frequency of occurrence
subsection_species <- data_pivotlong |>
  group_by(year,subsection,veg_type, no_plots, taxon_code) |>
  summarise(count = sum(presence), foo = count/no_plots) |>
  distinct()

subsection_species_wide <- subsection_species |>
  dplyr::select(-count,-no_plots) |>
  pivot_wider(names_from = taxon_code, values_from = foo, values_fill = 0)
```

```
# caluculating richness, evenness and diversity
subsection_indices <- subsection_species |>
  dplyr::group_by(year, subsection, veg_type) |>
  dplyr::summarise(
    # restrict to taxa that are present
    H = vegan::diversity(count[count > 0], index = "shannon"),
    S = dplyr::n_distinct(taxon_code[count > 0]),
    shannon = H,
```

```
    evenness = ifelse(S > 1, H / log(S), NA_real_),
    richness = S,
    .groups = "drop"
  )

years <- c("2007", "2012", "2017", "2022")
```

Data has been filtered to remove sections with only one remaining plot and subsections in saltmarsh, due to too few replicates in this vegetation type. Final calculates included 134 subsections (original dataset includes 140).

Richness

```
model_richness_sub_f <- lmer(
  richness ~ factor(year) + (1 | subsection),
  data = subsection_indices
)

summary(model_richness_sub_f)
```

```
Linear mixed model fit by REML ['lmerMod']
Formula: richness ~ factor(year) + (1 | subsection)
   Data: subsection_indices

REML criterion at convergence: 2159.7

Scaled residuals:
    Min      1Q  Median      3Q     Max
-5.0211 -0.4767 -0.0042  0.4437  5.2716

Random effects:
 Groups     Name        Variance Std.Dev.
 subsection (Intercept) 9.173    3.029
 Residual               2.219    1.490
Number of obs: 492, groups:  subsection, 134

Fixed effects:
                  Estimate Std. Error t value
(Intercept)      10.022586   0.292140  34.307
factor(year)2012  0.557353   0.185243   3.009
```

```
factor(year)2017 -0.753326   0.191972  -3.924
factor(year)2022  0.004956   0.195410   0.025

Correlation of Fixed Effects:
            (Intr) f()2012 f()2017
fctr(y)2012 -0.312
fctr(y)2017 -0.302  0.484
fctr(y)2022 -0.296  0.476   0.480
```

```
richness_emmeans_f <- emmeans(model_richness_sub_f, ~ year)

richness_pairs_f <- emmeans(model_richness_sub_f, ~ year) |>
  pairs()

richness_emmeans_f
```

```
 year emmean    SE  df lower.CL upper.CL
 2007  10.02 0.292 179     9.45    10.60
 2012  10.58 0.293 181    10.00    11.16
 2017   9.27 0.297 190     8.68     9.86
 2022  10.03 0.299 195     9.44    10.62

Degrees-of-freedom method: satterthwaite
Confidence level used: 0.95
```

```
richness_pairs_f
```

```
 contrast            estimate    SE  df t.ratio p.value
 year2007 - year2012 -0.55735 0.185 357  -3.009  0.0148
 year2007 - year2017  0.75333 0.192 359   3.924  0.0006
 year2007 - year2022 -0.00496 0.195 359  -0.025  1.0000
 year2012 - year2017  1.31068 0.192 358   6.838 <0.0001
 year2012 - year2022  0.55240 0.195 358   2.831  0.0251
 year2017 - year2022 -0.75828 0.198 356  -3.838  0.0008

Degrees-of-freedom method: satterthwaite
P value adjustment: tukey method for comparing a family of 4 estimates
```

```
richness_pairs_df <- as.data.frame(richness_pairs_f)
```

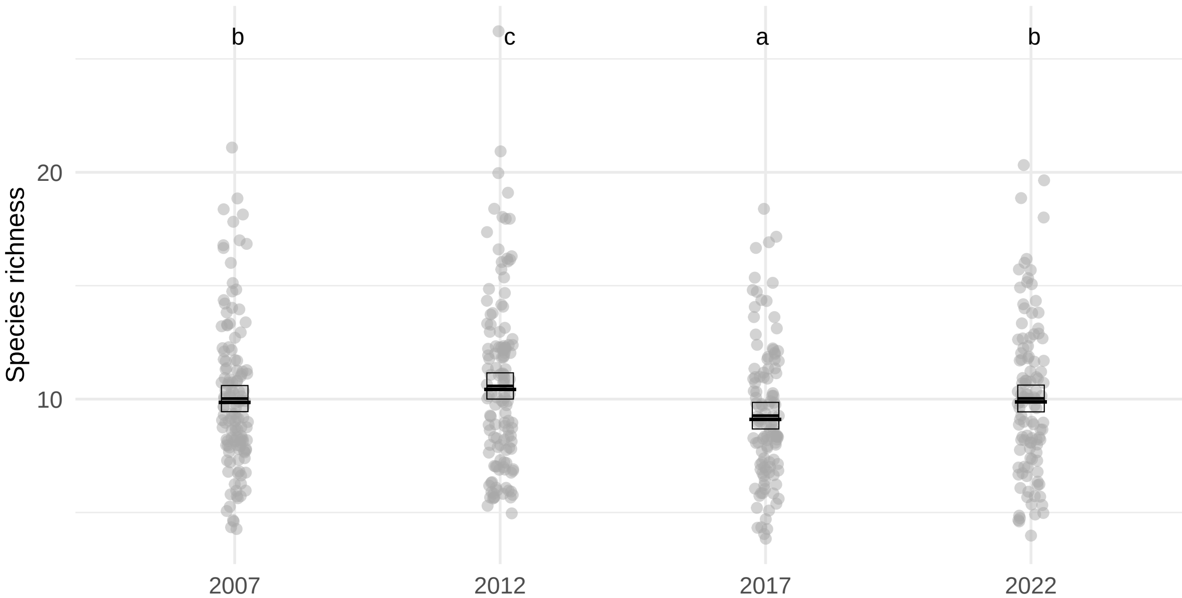

Figure 8: Species richness, number of species pr. subsection. Shared letters indicate non-significant differences among estimated means according to compact letter display.

Evenness

```
model_evenness_sub_f <- lmer(
  evenness ~ factor(year) + (1 | subsection),
  data = subsection_indices
)

summary(model_evenness_sub_f)
```

```
Linear mixed model fit by REML ['lmerMod']
Formula: evenness ~ factor(year) + (1 | subsection)
   Data: subsection_indices

REML criterion at convergence: -2413

Scaled residuals:
    Min      1Q  Median      3Q     Max 
-3.8146 -0.5175  0.0310  0.5727  2.9727 

Random effects:
```

```
 Groups     Name        Variance  Std.Dev.
 subsection (Intercept) 0.0001949 0.01396
 Residual               0.0002859 0.01691
Number of obs: 492, groups:  subsection, 134

Fixed effects:
                 Estimate Std. Error t value
(Intercept)      0.936409   0.001905 491.657
factor(year)2012 0.010617   0.002097   5.063
factor(year)2017 0.014378   0.002165   6.642
factor(year)2022 0.014919   0.002201   6.777

Correlation of Fixed Effects:
            (Intr) f()2012 f()2017
fctr(y)2012 -0.544
fctr(y)2017 -0.527  0.483
fctr(y)2022 -0.519  0.475   0.474
```

```
evenness_emmeans_f <- emmeans(model_evenness_sub_f, ~ year)

evenness_pairs_f <- emmeans(model_evenness_sub_f, ~ year) |>
  pairs()

evenness_emmeans_f
```

```
 year emmean      SE    df lower.CL upper.CL
 2007 0.9364 0.00190 330.2   0.9327   0.9402
 2012 0.9470 0.00192 332.8   0.9433   0.9508
 2017 0.9508 0.00199 355.7   0.9469   0.9547
 2022 0.9513 0.00203 368.5   0.9473   0.9553

Degrees-of-freedom method: satterthwaite
Confidence level used: 0.95
```

```
evenness_pairs_f
```

```
 contrast              estimate      SE  df t.ratio p.value
 year2007 - year2012 -0.010617 0.00210 352  -5.063 <0.0001
 year2007 - year2017 -0.014378 0.00216 358  -6.642 <0.0001
 year2007 - year2022 -0.014919 0.00220 360  -6.777 <0.0001
 year2012 - year2017 -0.003762 0.00217 355  -1.736  0.3065
```

```
 year2012 - year2022 -0.004303 0.00220 356  -1.953  0.2080
 year2017 - year2022 -0.000541 0.00224 351  -0.242  0.9950

Degrees-of-freedom method: satterthwaite
P value adjustment: tukey method for comparing a family of 4 estimates
```

```
evenness_pairs_df <- as.data.frame(evenness_pairs_f)
```

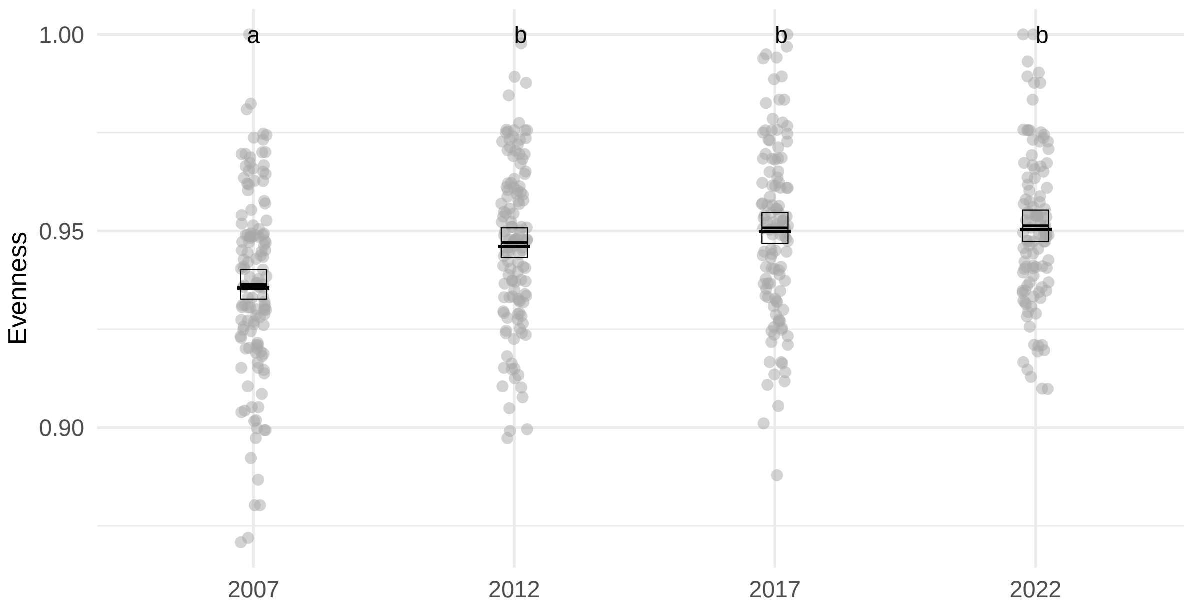

Figure 9: Evenness pr. subsection. Shared letters indicate non-significant differences among estimated means according to compact letter display.

Shannon

```
model_shannon_sub_f <- lmer(
  shannon ~ factor(year) + (1 | subsection),
  data = subsection_indices
)

summary(model_shannon_sub_f)
```

```
Linear mixed model fit by REML ['lmerMod']
Formula: shannon ~ factor(year) + (1 | subsection)
```

```
   Data: subsection_indices

REML criterion at convergence: -245.7

Scaled residuals:
    Min      1Q  Median      3Q     Max 
-4.7029 -0.4843  0.0323  0.5083  3.8910 

Random effects:
 Groups     Name        Variance Std.Dev.
 subsection (Intercept) 0.08371  0.2893  
 Residual               0.01481  0.1217  
Number of obs: 492, groups:  subsection, 134

Fixed effects:
                 Estimate Std. Error t value
(Intercept)       2.11191    0.02716  77.772
factor(year)2012  0.06457    0.01514   4.265
factor(year)2017 -0.04499    0.01569  -2.867
factor(year)2022  0.02751    0.01598   1.722

Correlation of Fixed Effects:
            (Intr) f()2012 f()2017
fctr(y)2012 -0.274                
fctr(y)2017 -0.265  0.484         
fctr(y)2022 -0.260  0.476   0.480 
```

```
shannon_emmeans_f <- emmeans(model_shannon_sub_f, ~ year)

shannon_pairs_f <- emmeans(model_shannon_sub_f, ~ year) |>
  pairs()

shannon_emmeans_f
```

```
 year emmean     SE  df lower.CL upper.CL
 2007   2.11 0.0272 167     2.06     2.17
 2012   2.18 0.0272 168     2.12     2.23
 2017   2.07 0.0275 175     2.01     2.12
 2022   2.14 0.0277 179     2.08     2.19

Degrees-of-freedom method: satterthwaite 
Confidence level used: 0.95 
```

```
shannon_pairs_f
```

```
 contrast             estimate     SE  df t.ratio p.value
 year2007 - year2012  -0.0646 0.0151 357  -4.265  0.0001
 year2007 - year2017   0.0450 0.0157 358   2.867  0.0227
 year2007 - year2022  -0.0275 0.0160 358  -1.722  0.3136
 year2012 - year2017   0.1096 0.0157 357   6.994 <0.0001
 year2012 - year2022   0.0371 0.0159 357   2.324  0.0945
 year2017 - year2022  -0.0725 0.0161 356  -4.491 <0.0001

Degrees-of-freedom method: satterthwaite
P value adjustment: tukey method for comparing a family of 4 estimates
```

```
shannon_pairs_df <- as.data.frame(shannon_pairs_f)
```

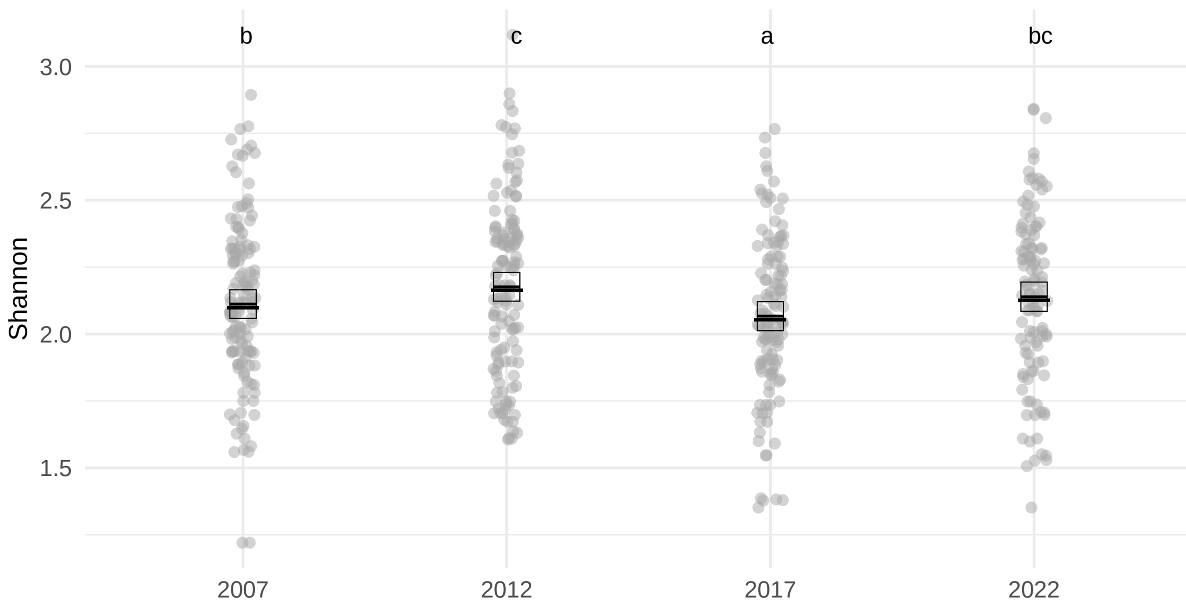

Figure 10: Shannon diversity pr. year calculate pr subsection. Shared letters indicate non-significant differences among estimated means according to compact letter display.

Turnover

```
# Check number of unique years per plot
plot_years <- subsection_species |>
```

```r
  group_by(subsection) |>
  summarise(n_years = n_distinct(year), .groups = "drop")

# Filter plots with at least 2 years (turnover requires temporal comparison)
valid_plots <- plot_years |>
  filter(n_years > 1) |>
  pull(subsection)

# Filter data to valid plots only
species_long_filtered <- subsection_species |>
  filter(subsection %in% valid_plots)

# Calculate turnover per plot between years
turnover_results <- turnover(
  df = species_long_filtered,
  time.var = "year",
  species.var = "taxon_code",
  abundance.var = "foo",
  replicate.var = "subsection",
  metric = "total"
)
```

```r
m_turnover_sub_f <- lmer(total ~ factor(year) + (1|subsection),
                         data = turnover_results)

summary(m_turnover_sub_f)
```

```
Linear mixed model fit by REML ['lmerMod']
Formula: total ~ factor(year) + (1 | subsection)
   Data: turnover_results

REML criterion at convergence: -451.7

Scaled residuals:
     Min       1Q   Median       3Q      Max
-2.11602 -0.58928  0.01139  0.56350  2.53936

Random effects:
 Groups     Name        Variance Std.Dev.
 subsection (Intercept) 0.008066 0.08981
 Residual               0.010461 0.10228
Number of obs: 358, groups:  subsection, 131
```

```
Fixed effects:
                 Estimate Std. Error t value
(Intercept)       0.20204    0.01199  16.844
factor(year)2017  0.03189    0.01323   2.412
factor(year)2022 -0.04771    0.01346  -3.544

Correlation of Fixed Effects:
            (Intr) f()201
fctr(y)2017 -0.520
fctr(y)2022 -0.511  0.483
```

```r
turnover_emmeans <- emmeans(m_turnover_sub_f, ~ year) |>
  pairs()

turnover_emmeans
```

```
 contrast            estimate     SE  df t.ratio p.value
 year2012 - year2017  -0.0319 0.0132 238  -2.412  0.0438
 year2012 - year2022   0.0477 0.0135 240   3.544  0.0014
 year2017 - year2022   0.0796 0.0136 231   5.869 <0.0001

Degrees-of-freedom method: satterthwaite
P value adjustment: tukey method for comparing a family of 3 estimates
```

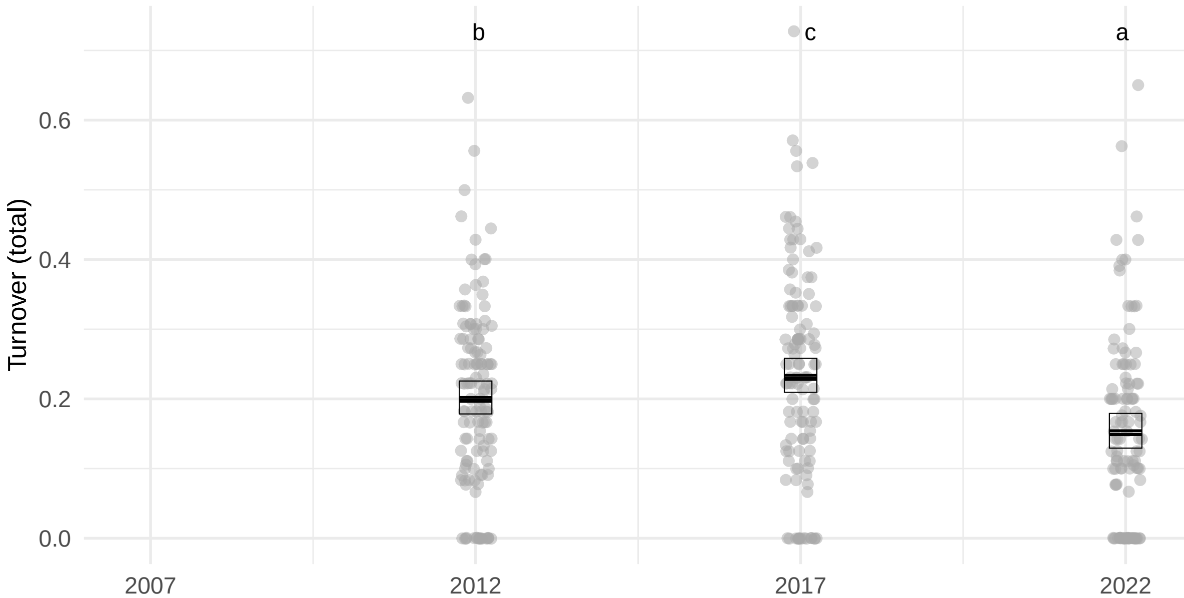

Figure 11: Turnover pr. year calculate pr subsection, but only pr present all survey years. A totalt of 131 subsections maintain more than 1 plots for all 4 surveys. Shared letters indicate non-significant differences among estimated means according to compact letter display.

Plots combined

```
Warning: Removed 1 row containing missing values or values outside the scale range
(`geom_crossbar()`).
```

```
Warning: Removed 1 row containing missing values or values outside the scale range
(`geom_point()`).
```

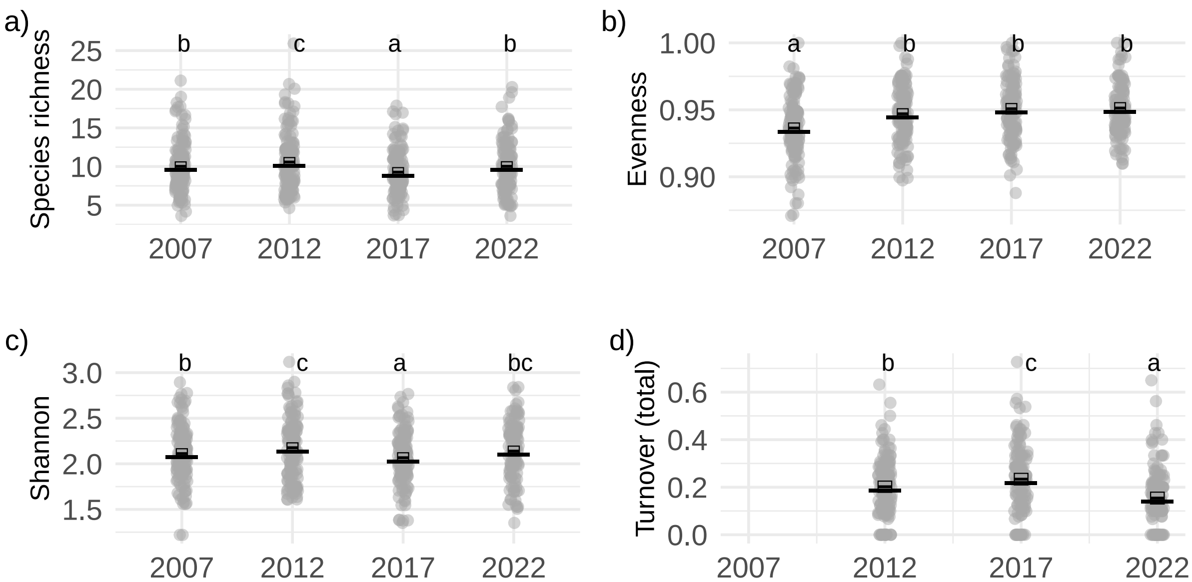

Figure 12: Diversity metrices, a) Shannon diversity, b) Species richness, c) Evenness, c) Species turnover. Shared letters indicate non-significant differences among estimated means according to compact letter display.

3-2 Community composition

```
community_matrix <- subsection_species_wide |>
  unite("sub_year_vt", subsection, year, veg_type, sep = "_", remove = FALSE) |>
  as.data.frame() |>
  column_to_rownames(var = "sub_year_vt") |>
  dplyr::select(-year, -subsection, -veg_type, -no_plots)

meta <- subsection_species_wide |>
  mutate(sub_year_vt = paste(subsection, year, veg_type, sep = "_")) |>
  dplyr::select(sub_year_vt, year, veg_type, subsection) |>
  distinct()
```

NMDS

Community composition data ready for analysis consists of 134 subsections from 4 years of surveys. 99 species have been observed across all surveys.

```
Call:
metaMDS(comm = community_matrix, k = 3, trymax = 100)

global Multidimensional Scaling using monoMDS

Data:     community_matrix
Distance: bray

Dimensions: 3
Stress:     0.1317335
Stress type 1, weak ties
Best solution was repeated 1 time in 20 tries
The best solution was from try 20 (random start)
Scaling: centring, PC rotation, halfchange scaling
Species: expanded scores based on 'community_matrix'
```

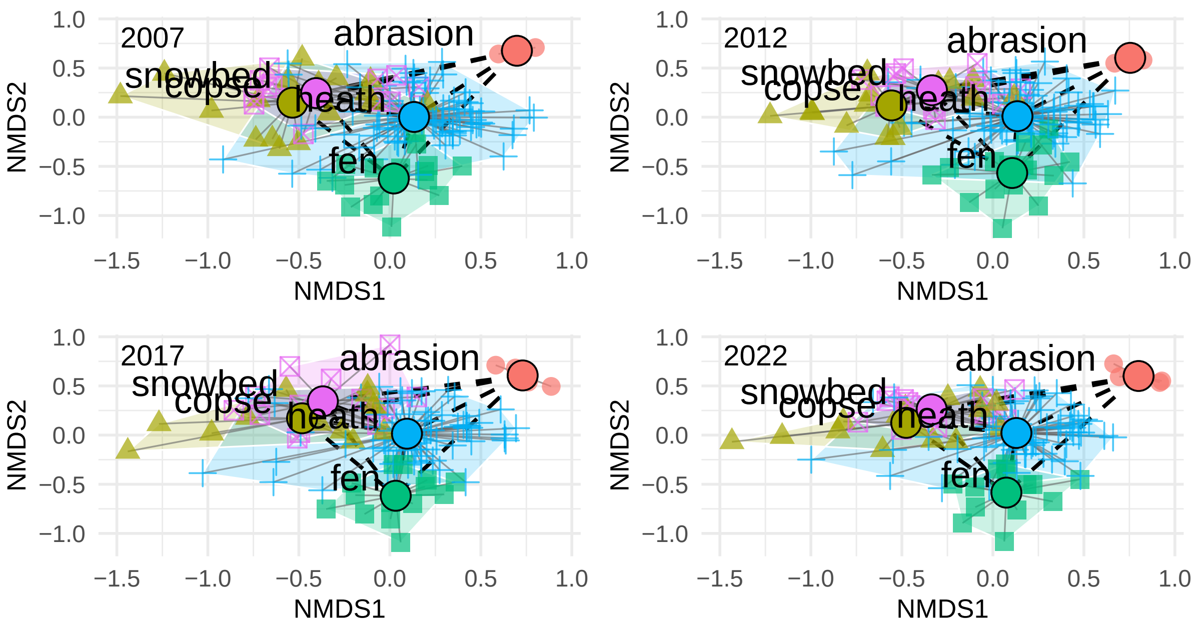

Figure 13: Plant community dynamics. Non-metric multidimensional scaling (NMDS) ordination of vegetation composition across years and vegetation types. Each panel represents subset of years of the full NMDS including all data to keep NMDS distances comparable. Grey lines connect plots within vegetation types, the within-type centroid distances. Black dashed lines indicate distances between vegetation-type centroids.

Between vegetation composition

`Model: dist ~ factor(year) + (1 | veg_pair), fitted by REML`

`N observations: 40 | N groups (veg_pair): 10`

Table 1: Fixed effects: centroid distance ~ survey year (factor)

term	estimate	std.error	statistic
(Intercept)	0.886	0.119	7.462
Year 2012	-0.014	0.017	-0.780
Year 2017	-0.018	0.017	-1.024
Year 2022	-0.025	0.017	-1.448

Table 2: Estimated mean centroid distances per survey

year	emmean	SE	lower.CL	upper.CL
2007	0.886	0.119	0.618	1.154
2012	0.873	0.119	0.605	1.140
2017	0.868	0.119	0.600	1.136
2022	0.861	0.119	0.593	1.129

Table 3: Pairwise contrasts between surveys (Sidak-adjusted)

contrast	estimate	SE	t.ratio	p.value
year2007 - year2012	0.014	0.017	0.780	0.970
year2007 - year2017	0.018	0.017	1.024	0.897
year2007 - year2022	0.025	0.017	1.448	0.647
year2012 - year2017	0.004	0.017	0.244	1.000
year2012 - year2022	0.012	0.017	0.668	0.986
year2017 - year2022	0.007	0.017	0.424	0.999

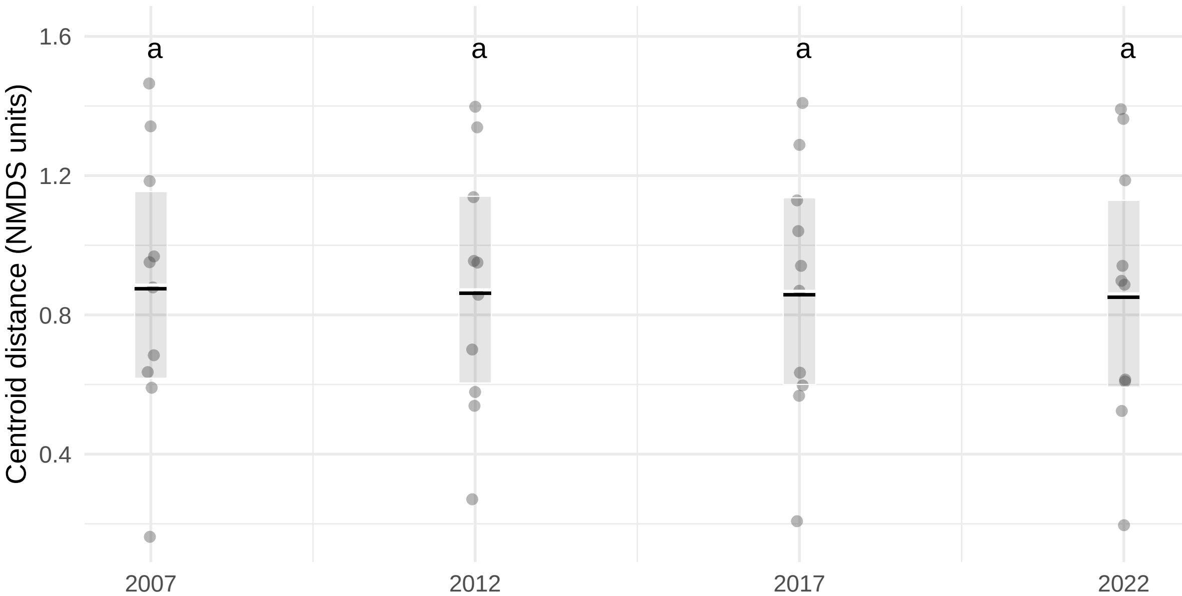

Figure 14: Between plant community dynamics. Estimated mean distance between vegetation-type centroids within each survey year based on linear mixed-effects models. Black points show estimated means (± SE), and grey dots represent raw subsection-level distances (in NMDS units). Shared letters indicate non-significant differences among estimated means according to compact letter display.

Within vegetation composition

3-4 Functional types

Functional groups

Specific species

4 Phenology (discussion)

Data available from https://data.g-e-m.dk (GEM 2026b, 2026c).

4-1 NDVI

```
Computing bootstrap confidence intervals ...
```

```
6 warning(s): Model failed to converge with max|grad| = 0.00203777 (tol = 0.002, component 1)
  See ?lme4::convergence and ?lme4::troubleshooting. (and others)

Computing bootstrap confidence intervals ...


4 warning(s): Model failed to converge with max|grad| = 0.00206721 (tol = 0.002, component 1)
  See ?lme4::convergence and ?lme4::troubleshooting. (and others)
```

Table 4: Fixed effects

term	estimate	std.error	statistic	lower	upper
(Intercept)	0.4134	0.0348	11.8820	0.3421	0.4823
year_c	0.0010	0.0004	2.6589	0.0002	0.0018

Table 5: Random effects (variance components)

group	term	estimate
section:plot_id	sd__(Intercept)	0.0385
plot_id	sd__(Intercept)	0.1544
Residual	sd__Observation	0.0677

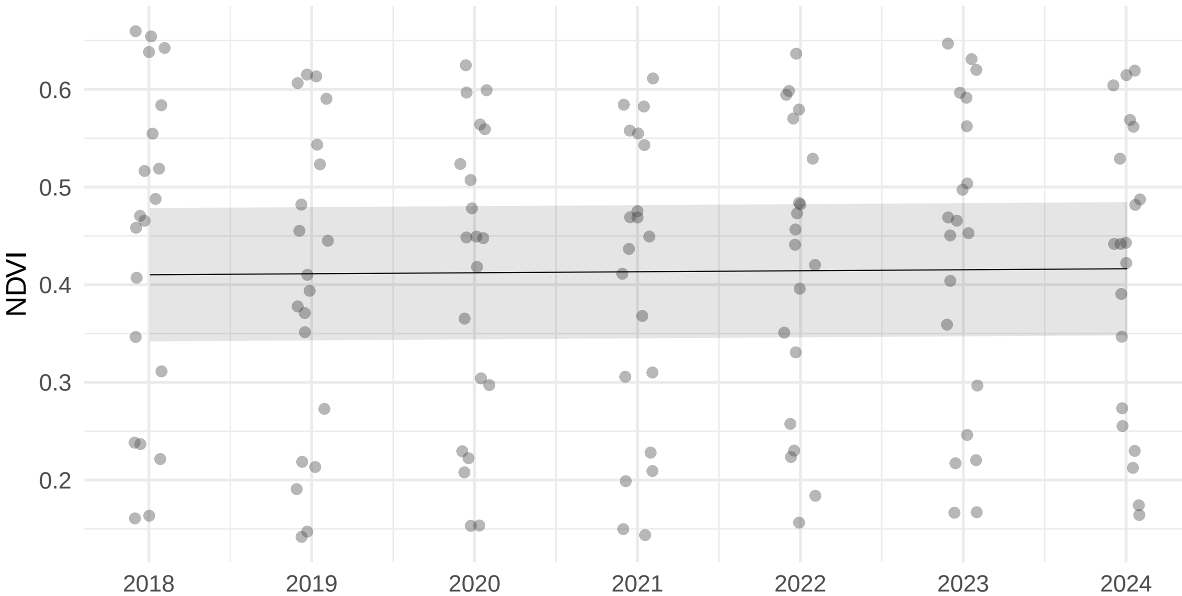

Figure 15: NDVI change over time. Mean NDVI per plot per year, based on in situ measurements from 2018 to 2024. The trend line shows a significant increase of 0.001 NDVI units per year (95% CI: 3e-04–0.0018), representing a cumulative increase of 0.0061 NDVI units over the study period.

4-2 Salix reproductive phenology

```
Note: adjust = "tukey" was changed to "sidak"
because "tukey" is only appropriate for one set of pairwise comparisons
```

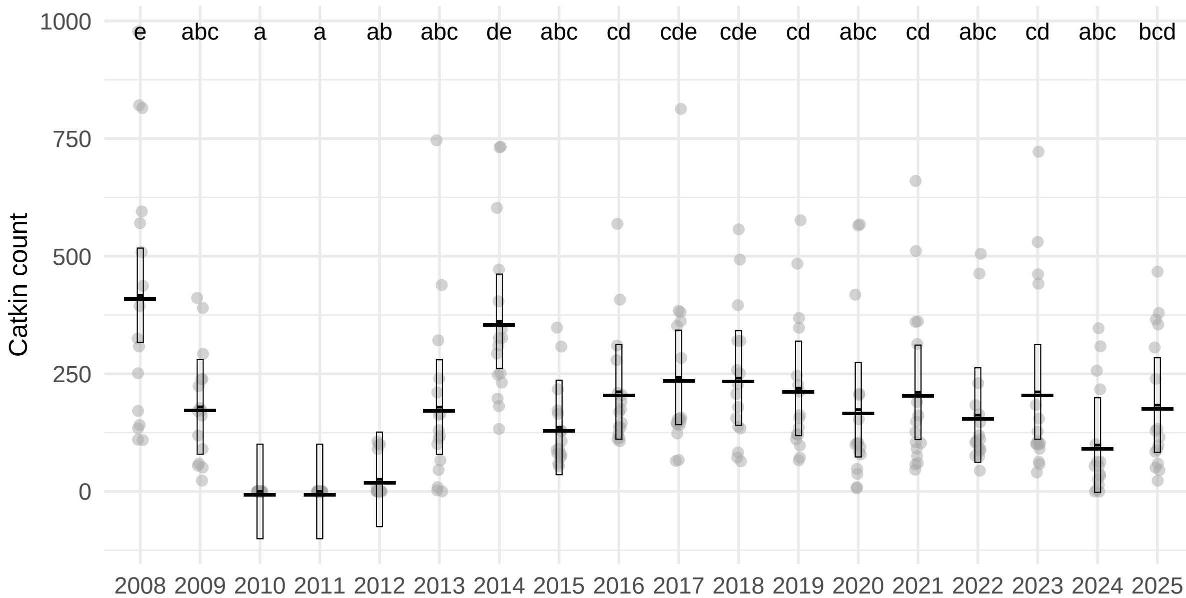

Figure 16: Phenology

5 References

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